

Curriculum Vitae

Yehui Hou, Ph.D. ✉ yehuihou@sjtu.edu.cn ☎ +86 13121275661

🌐 <https://inspirehep.net/authors/2052049?ui-citation-summary=true>

👤 <https://orcid.org/0000-0002-9434-3930>



Personal Information

Date of Birth: April 1997 Residence: Haidian District, Beijing Highest Degree: Ph.D.

Professional Experience

Sept 2025 – June 2027 **Tsung-Dao Lee Institute, Shanghai Jiao Tong University** TDLI Fellowship
Co-supervisor: *Prof. Dong Lai* ✉ donglai@sjtu.edu.cn
Co-supervisor: *Prof. Yosuke Mizuno* ✉ mizuno@sjtu.edu.cn

Education

Sept 2019 – June 2025 **Peking University** Ph.D. in Theoretical Physics
Ph.D Advisor: *Prof. Bin Chen* ✉ chenbin1@nbu.edu.cn
Sept 2015 – June 2019 **Beihang University** B.Sc. in Applied Physics

Major Honors

2025 **Super Postdoctoral Fellowship** Tsung-Dao Lee Institute, Shanghai Jiao Tong University
2024 **National Scholarship; Academic Innovation Award** Peking University
2023 **Presidential Scholarship** Peking University
2021 **Special Scholarship for Academic Master's Students; Outstanding Research Award** Peking University

Teaching and Research Skills

Fields of Expertise: Electrodynamics, Quantum Mechanics and other core courses in theoretical physics, General Relativity, Relativistic Astrophysics ...
Teaching Experience: Teaching Assistant for General Relativity (Prof. Bin Chen, Peking University, Spring 2021)
Programming Skills: Wolfram Mathematica, Python, $\LaTeX$, ...

Research Interests

My research interests focus on black hole physics and relativistic astrophysics, including but not limited to: probing physical processes in strong-gravity regions and the fundamental nature of gravity through black hole imaging; investigating accretion disks, jets, and electromagnetic field structures; mechanisms of black

hole energy extraction; black hole perturbations and strong-field gravitational waves; electrodynamics of compact objects; and particle dynamics in curved spacetime.

Research Output (Past Five Years)

Over the past five years, I have completed **25** publications, including **8** as first author, **3** as co-first author (listed alphabetically), and **6** as corresponding author. Among them, **20** papers have been published in leading journals in the field, including *The Astrophysical Journal Letters*, *Physical Review D*, *Journal of Cosmology and Astroparticle Physics*, and *Journal of High Energy Physics*. To date, my publications have received over **790** citations (according to INSPIRE).

- 1 **Yehui Hou**, J. Huang, and B. Chen, “Distinct Near-Horizon Trend of Synchrotron Polarization in Kerr Spacetime,” Jun. 2026. arXiv: 2606.19229 [gr-qc].
- 2 J. Huang, **Yehui Hou**, Z. Zhong, M. Guo, and B. Chen, “Black Hole Ringdown Seen in Photon Polarization Swings,” May 2026. arXiv: 2605.11499 [astro-ph.HE].
- 3 **Yehui Hou**, J. Huang, Y. Mizuno, M. Guo, and B. Chen, “Near-horizon Polarization as a Diagnostic of Black Hole Spacetime,” *Astrophys. J. Lett.*, vol. 988, no. 2, p. L51, 2025.  DOI: 10.3847/2041-8213/adee09. arXiv: 2409.07248 [gr-qc].
- 4 **Yehui Hou**, Z. Zhang, M. Guo, and B. Chen, “A new analytical model of magnetofluids surrounding rotating black holes,” *JCAP*, vol. 02, p. 030, 2024.  DOI: 10.1088/1475-7516/2024/02/030. arXiv: 2309.13304 [gr-qc].
- 5 **Yehui Hou**, Z. Zhang, M. Guo, and B. Chen, “Electromagnetic effects on charged particles in a near-horizon-extreme-Kerr geometry,” *Phys. Rev. D*, vol. 107, no. 12, p. 124 014, 2023.  DOI: 10.1103/PhysRevD.107.124014. arXiv: 2301.08467 [gr-qc].
- 6 **Yehui Hou**, Z. Zhang, H. Yan, M. Guo, and B. Chen, “Image of a Kerr-Melvin black hole with a thin accretion disk,” *Phys. Rev. D*, vol. 106, no. 6, p. 064 058, 2022.  DOI: 10.1103/PhysRevD.106.064058. arXiv: 2206.13744 [gr-qc].
- 7 **Yehui Hou**, P. Liu, M. Guo, H. Yan, and B. Chen, “Multi-level images around Kerr–Newman black holes,” *Class. Quant. Grav.*, vol. 39, no. 19, p. 194 001, 2022.  DOI: 10.1088/1361-6382/ac8860. arXiv: 2203.02755 [gr-qc].
- 8 **Yehui Hou**, M. Guo, and B. Chen, “Revisiting the shadow of braneworld black holes,” *Phys. Rev. D*, vol. 104, no. 2, p. 024 001, 2021.  DOI: 10.1103/PhysRevD.104.024001. arXiv: 2103.04369 [gr-qc].
- 9 B. Chen, **Yehui Hou**, Y. Song, and Z. Zhang, “Polarization patterns of the hot spots plunging into a Kerr black hole,” *Phys. Rev. D*, vol. 111, no. 8, p. 083 045, 2025.  DOI: 10.1103/PhysRevD.111.083045. arXiv: 2407.14897 [astro-ph.HE].
- 10 B. Chen, **Yehui Hou**, J. Li, and Y. Shen, “Energy extraction from a Kerr black hole via magnetic reconnection within the plunging region,” *Phys. Rev. D*, vol. 110, no. 6, p. 063 003, 2024.  DOI: 10.1103/PhysRevD.110.063003. arXiv: 2405.11488 [gr-qc].
- 11 B. Chen, **Yehui Hou**, and Z. Hu, “On emergent conformal symmetry near the photon ring,” *JHEP*, vol. 05, p. 115, 2023.  DOI: 10.1007/JHEP05(2023)115. arXiv: 2212.02958 [hep-th].
- 12 Z. Zhang, **Yehui Hou**, Y. Song, Y. Mizuno, and B. Chen, “Polarization Architecture of Steady GRMHD Jets from the Horizon to Infinity,” Jul. 2026. arXiv: 2607.01820 [astro-ph.HE].

- 13 F. Zhou, J. Huang, Y. Li, *et al.*, “Nonthermal Synchrotron Emission and Polarization Signatures during Black Hole Flux Eruptions,” *Astrophys. J.*, vol. 1002, no. 2, p. 152, 2026.  DOI: 10.3847/1538-4357/ae5e67. arXiv: 2512.06803 [astro-ph.HE].
- 14 Q. Wan, **Yehui Hou**, and M. Guo, “Probing the scalar hair of rotating Horndeski black holes through thick disk images,” *Phys. Rev. D*, vol. 113, no. 8, p. 083 023, 2026.  DOI: 10.1103/zjtb-3dyz. arXiv: 2512.00917 [gr-qc].
- 15 Y. Song, **Yehui Hou**, L. Huang, and B. Chen, “Hybrid black hole- and disk-driven jets: Steady axisymmetric ideal MHD modeling,” *Phys. Rev. D*, vol. 113, no. 4, p. 043 054, 2026.  DOI: 10.1103/kv5f-kwzm. arXiv: 2507.23281 [astro-ph.HE].
- 16 Q. Wan, **Yehui Hou**, Y. Huang, P.-C. Li, M. Guo, and B. Chen, “Distorting Kerr images with parity-odd scalar hair,” *Phys. Rev. D*, vol. 114, no. 8, p. 084 012, 2026.  DOI: 10.1103/f3wb-bv6m. arXiv: 2605.28376 [gr-qc].
- 17 X. Wang, **Yehui Hou**, X. Wan, M. Guo, and B. Chen, “Geodesics and shadows in the Kerr-Bertotti-Robinson black hole spacetime,” *JCAP*, vol. 02, p. 050, 2026.  DOI: 10.1088/1475-7516/2026/02/050. arXiv: 2507.22494 [gr-qc].
- 18 X. Wan, Z. Zhang, F.-S. Wei, **Yehui Hou**, and B. Chen, “Critical behavior of photon rings in Kerr-Bertotti-Robinson spacetime,” *Phys. Rev. D*, vol. 114, no. 4, p. 044 005, 2026.  DOI: 10.1103/fyky-bkbg. arXiv: 2603.25049 [gr-qc].
- 19 Z. Zhang, **Yehui Hou**, M. Guo, Y. Mizuno, and B. Chen, “Autocorrelation signatures in time-resolved black hole flare images: Secondary peaks and convergence structure,” *Phys. Rev. D*, vol. 112, no. 8, p. 083 024, 2025.  DOI: 10.1103/zmnz-p2rs. arXiv: 2503.17200 [astro-ph.HE].
- 20 Z. Zhang, **Yehui Hou**, M. Guo, and B. Chen, “Imaging thick accretion disks and jets surrounding black holes,” *JCAP*, vol. 05, p. 032, 2024.  DOI: 10.1088/1475-7516/2024/05/032. arXiv: 2401.14794 [astro-ph.HE].
- 21 Z. Zhang, **Yehui Hou**, Z. Hu, M. Guo, and B. Chen, “Polarized images of charged particles in vortical motions around a magnetized Kerr black hole,” *JCAP*, vol. 03, p. 013, 2024.  DOI: 10.1088/1475-7516/2024/03/013. arXiv: 2304.03642 [gr-qc].
- 22 Z. Zhang, **Yehui Hou**, and M. Guo, “Observational signatures of rotating black holes in the semiclassical gravity with trace anomaly*,” *Chin. Phys. C*, vol. 48, no. 8, p. 085 106, 2024.  DOI: 10.1088/1674-1137/ad432b. arXiv: 2305.14924 [gr-qc].
- 23 X. Wang, **Yehui Hou**, and M. Guo, “How different are shadows of compact objects with and without horizons?” *JCAP*, vol. 05, p. 036, 2023.  DOI: 10.1088/1475-7516/2023/05/036. arXiv: 2301.04851 [gr-qc].
- 24 Z. Hu, **Yehui Hou**, H. Yan, M. Guo, and B. Chen, “Polarized images of synchrotron radiations in curved spacetime,” *Eur. Phys. J. C*, vol. 82, no. 12, p. 1166, 2022.  DOI: 10.1140/epjc/s10052-022-11144-9. arXiv: 2203.02908 [gr-qc].
- 25 Y. Shen, **Yehui Hou**, Z.-Y. Fan, M. Guo, and B. Chen, “Imaging a Semi-Analytical Jet model Generated by 3D GRMHD Simulation,” Nov. 2023. arXiv: 2311.14954 [astro-ph.HE].

Academic Activities

Conferences

- 2022.7 **2022 Annual Meeting of the Division of Gravitation and Relativistic Astrophysics, Chinese Physical Society, Jiangsu, China**

Academic Activities (continued)

- 2023.4 **2023 Annual Meeting of the Division of Gravitation and Relativistic Astrophysics, Chinese Physical Society**, Chongqing, China
- 2023.12 **2023 Black Hole Imaging Workshop**, Beijing, China
- 2024.7 **“New Horizons for Psi” Workshop**, Lisbon, Portugal
- 2024.11 **2024 Black Hole Imaging Workshop**, Hunan, China
- 2025.4 **2025 Annual Meeting of the Division of Gravitation and Relativistic Astrophysics, Chinese Physical Society**, Yunnan, China
- 2025.11 **2025 Black Hole Imaging Workshop**, Jiangsu, China
- 2026.4 **2026 Annual Meeting of the Division of Gravitation and Relativistic Astrophysics, Chinese Physical Society**, Hainan, China
- 2026.7 **CosPA2026**, Christchurch, New Zealand
- 2026 Black Hole Imaging Workshop**, Gansu, China
- 2026.8 **ICGAC16**, Guangdong, China
- 2nd “New Horizons for Psi” Workshop**, Beijing, China

Contributed Talks

- 2022.7 **"Image of a Kerr-Melvin Black Hole with a Thin Accretion Disk"**.
2022 Annual Meeting of the Division of Gravitation and Relativistic Astrophysics, Chinese Physical Society, hosted by the Institute of Theoretical Physics, Chinese Academy of Sciences
- 2023.4 **"Electromagnetic Effects on Charged Particles in a NHEK Geometry"**.
2023 Annual Meeting of the Division of Gravitation and Relativistic Astrophysics, Chinese Physical Society, hosted by Chongqing University
- 2023.12 **"A New Analytical Model of Magnetofluids Surrounding Rotating Black Holes"**.
2023 Black Hole Imaging Workshop, hosted by the Center for High Energy Physics, Peking University
- 2024.9 **"Unique Imprint of Black Hole Spin on Near-Horizon Polarization"**.
Invited visit and seminar, Shanghai Astronomical Observatory, Chinese Academy of Sciences
- 2024.11 **"Unique Imprint of Black Hole Spin on Near-Horizon Polarization"**.
2024 Black Hole Imaging Workshop, hosted by the College of Physics and Electronic Science, Hunan Normal University
- 2025.4 **"Analytical Modeling and Imaging Study of Horizon-Scale Accretion Flows around Kerr Black Holes"**.
2025 Annual Meeting of the Division of Gravitation and Relativistic Astrophysics, Chinese Physical Society, hosted by the Faculty of Science, Kunming University of Science and Technology
- 2025.11 **"Black Hole-Disk Driven Jet Outflows: Steady Axisymmetric GRMHD Modeling"**.
2025 Black Hole Imaging Workshop, hosted by the College of Physics Science and Technology, Yangzhou University
- 2026.7 **"Kerr Black Hole Ringdown Seen in Photon Polarization Swings"**.
CosPA2026, hosted by the School of Physical and Chemical Sciences, University of Canterbury
- "Polarization Architecture of Steady GRMHD Jets from the Event Horizon to Infinity"**.
2026 Black Hole Imaging Workshop, hosted by the School of Physical Science and Technology, Lanzhou University
- 2026.8 **"Black Hole Ringdown Seen in Photon Polarization Swings"**.
ICGAC16, hosted by the TianQin Research Center, Sun Yat-sen University

Academic Activities (continued)

"Black Hole Ringdown Seen in Photon Polarization Swings".

2nd "New Horizons for Psi" Workshop, hosted by the International Centre for Theoretical Physics Asia-Pacific